

GAURAV SINGH BHATI

Udaipur, Rajasthan | gauravbhati2099@gmail.com | Portfolio | LinkedIn | GitHub | +91-7425841804

Summary

B.Tech Computer Science candidate and Software Engineer focused on **Python, C++, DSA, AI/ML, backend engineering, and workflow automation**. Built **8+ AI workflows** and backend systems spanning FastAPI services, agentic AI, RAG, SAP-integrated automation, and full-stack applications using **LangGraph, LangChain, SQL/NoSQL, REST/Graph APIs, AWS, Docker, Linux, and React**.

Education

B.Tech, Computer Science & Engineering

Expected Aug 2027

Geetanjali Institute of Technical Studies, Udaipur

Technical Skills

Languages / Web: Python, C++, SQL, JavaScript, HTML, CSS, React

DSA: Data Structures, Algorithms, Complexity Analysis, Arrays, Strings, Hashing, Linked Lists, Stacks, Queues, Trees, Heaps, Graphs, Sorting, Searching, Recursion, Greedy, Dynamic Programming

AI/ML: Generative AI, LLMs, Agentic AI, RAG, LangChain, LangGraph, Prompt Engineering, NLP, Semantic Search, Embeddings, Ollama, Gemini, OpenAI APIs

Backend: FastAPI, REST APIs, Microsoft Graph API, Async Processing, Pydantic, Authentication, Background Workers

Data / Enterprise / DevOps: PostgreSQL, MongoDB, Redis, FAISS, Schema Design, Query Optimization, SAP, SAP OData/API, ABAP Fundamentals, MRP, Quality Management, Linux, AWS, Docker, Git, GitHub, CI/CD, OpenTelemetry

Experience

Workflow & Automation Intern

Jul 2026 – Present

Tempens Instruments India Ltd.

Udaipur, Rajasthan

- Automated **8 SRM forms** with SharePoint and Power Automate, centralizing **form submission, ticketing, approval, notification, and status-tracking workflows** for a platform handling **10 forms, 100+ tickets, and 70+ approvals** across multiple plants.
- Transformed the prototype into a **centralized SRM platform** with SQL-backed workflow state and layered **user authentication and role-based authorization**, integrating Python, LangGraph, SQL, REST APIs, Microsoft Graph API, and SAP integration across forms, tickets, approvals, and notifications.
- Extended **7 business processes** with SMS-based customer communication alongside existing internal/client email notifications, integrating reusable templates with the existing SMS API across **quotation, invoice, email-enquiry, approval, notification, and SMS workflows**.
- Supported SAP OData/API processes across **MRP, Quality Management, and user administration**, and created SOPs and technical guides for implemented workflows and operational handover.

AI Engineer Intern

Jun 2026 – Jul 2026

Onepws Pvt. Ltd.

- Optimized a Python/FastAPI AI platform with **5+ workflows** including **natural-language query, memory, and AI-processing workflows**, cutting query-generation latency from **~30s to <5s** and memory retrieval from **~2.3s to <100ms** through schema-aware query planning and retrieval optimization.
- Delivered an image-generation service for a **~10-person team** with layered **user authentication, role-based access, and usage controls** around a shared OpenAI account; deployed services on AWS/Linux with Docker, Git, REST APIs, process management, and CI/CD.

- Researched and compared **5 competing healthcare products** and tested MedLipi with **100+ medical prescription cases**, evaluating product behavior, workflows, edge cases, and practical healthcare use cases.
- Collaborated with the research team to analyze MedLipi's portable patient-history workflows, studying data structure, software behavior, and interoperability against **HIPAA considerations and SOAP protocols**.

Selected Projects

Intelligent Customer Support Platform Python / AI Backend

- Built an AI backend supporting **8+ workflows** including **conversation, memory, retrieval, scheduling, email, document, and business-automation workflows** using Python, FastAPI, LangGraph, MongoDB, PostgreSQL, and Redis, with layered **user authentication and role-based authorization**.
- Engineered memory **extraction, ranking, retrieval, deduplication, and expiry**, enabling persistent conversational context with controlled memory growth.

RAG Policy Assistant Python / RAG

- Built a local enterprise RAG pipeline over **20+ policy documents** using Python, FastAPI, LangChain, FAISS, Ollama, and OCR, with an extensible option to add additional context for more policy-specific responses.

Natural Language MongoDB Query Engine Python / LLM

- Built a schema-aware **natural-language MongoDB query workflow** converting requests to MongoDB operations via runtime schema discovery and LLM-assisted query planning, with layered **user authentication and role-based authorization** before data access.

Certifications

RHCSA – Red Hat Certified System Administrator | **Machine Learning Specialization** – DeepLearning.AI | **Google GenAI Intensive**